MILITARY SPECIFICATION

BURSTER, PROJECTILE, M21, LOADING ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing of one burster designated as Burster, Projectile, M21.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607- Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1310

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DRAWINGS

ARMY

9238782 - Projectile, Burster, M21, Assembly
9242137 - Box, Fiber, Packing, Ammunition for Burster, Charge M21
9242138 - Box, Packing, Ammunition for Burster Charge M21

PUBLICATIONS

ARMY

ET-9238782 - Equipment Tabulation

(Copies of specification, standards, drawings and publications required by contractors in connection with specific procurement and functions should be obtained from the procuring activity or ad directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Material shall be in accordance with applicable drawings and specifications.

3.2 Burster.-The burster shall comply with all requirements specified on drawing (dwg) 9238782 and with all requirements specified in applicable specifications.

3.3 Workmanship

3.3.1 All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, cracks, surface defects, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the part be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the

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supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn

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and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Pellets of one lot interfix number from one manufacturer.
b. Parts of one lot interfix number from one manufacturer.

4.2.2 Examination.-Inspection for Critical defects (and Major defects, when so specified) shall be 100 precent. Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL-STD-105 except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.3.

4.2.2.1 Tube, burster, prior to assembly (see dwg. 9238783 covering a detail of dwg. 9238782).

Categories	Defect	Method of Inspection	Code No.
Critical: None Defined			
Major: AQL 0.65 percent			
101.	Total length	Gage	01001
102.	Inside diameter	Gage	01002
103.	Wall thickness	Gage	01003
104.	Tube straightness	Gage	01004
Minor: AQL 1.0 percent			

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201.	Finish improper or missing (where applicable	Visual	01005
202.	Evidence of poor workmanship (see 3.3)	Visual	01006

4.2.2.2 Assembly, prior to securing discs (see dwg. 9238782).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Explosive above flush	Gage	02001
2.	Surface of explosive more than .01 below end of tube	Gage	02002
Major:	None defined		
Minor:	None defined		

4.2.2.3 Assembly (see dwg. 9238782).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Burster tube cracked	Visual	03001
2.	Charge loose in tube	Manual	03002
3.	Closing disc missing	Visual	03003
Major:	AQL 0.40 percent		
101.	Total length, maximum (max)	Gage	03004
102.	Chamber gage	Gage	03005
Minor:	AQL 0.65 percent		
201.	Closing disc torn or improperly sealed	Visual	03005
202.	Coating damaged, bare area except slight scratches (where applicable)	Visual	03006
203.	Evidence of poor workmanship (see 3.3)	Visual	03007

4.2.2.4 Carton prior to sealing (see dwg. 9242137)

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major: AQL 0.40 percent

101.	Burster projectile missing.....	Visual	04001

Minor: AQL 0.65 percent

201.	Any packing component missing.....	Visual	04002

4.2.2.5 Sealed carton (see dwg. 9242137)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Tape improperly applied.....	Visual	05001
Minor:	AQL 1.00 percent		
201.	Marking misleading or unidentifiable.....	Visual	05002
202.	Contents loose upon shaking.....	Manual	05003

4.2.2.6 Sealed wood packing box (see dwg. 9242138)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Box damaged.....	Visual	06001
102.	Strapping missing, broken or loose.....	Visual Manual	06002
Minor:	AQL 1.50 percent		
201.	Strapping improperly assembled.....	Visual	06003
202.	Lid improperly assembled.....	Visual	06004
203.	Markings misleading or unidentifiable.....	Visual	06005
204.	Contents loose in box.....	Manual	06006

4.2.3 Inspection Equipment - Equipment Tabulation Number ET-9238782 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

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4.2.3.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.3.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.2.3.3 Government use of contractor's inspection equipment. The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-The M21 projectile burster shall be preserved and packaged in accordance with dwg. 9242137.

5.2 Packing

5.2.1 Level C.-The cartons shall be packed in accordance with dwg. 9242138.

5.3 Marking.-The cartons and boxes shall be marked in accordance with dwg. 9242137 or 9242138 as applicable.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all informaton necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1310-A100

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